Explainable and Age-Aware Learning Analytics for Teacher-Facing Systems: A Synthesis and Position Paper

Saba Zafar

Abstract

Explainable artificial intelligence (XAI) and learning analytics have matured rapidly in education research over the past five years, but almost entirely around predicting and explaining student outcomes. This paper synthesizes recent peer-reviewed literature, 2021 to 2026, on three converging threads: explainable AI in educational analytics, teacher technology adoption and its relationship to age, and trait- or state-aware personalization in learning analytics. This synthesis finds that teacher-facing explainable AI is an emerging but still narrow subfield, in which teachers typically act as expert validators of explanations concerning students rather than as the subject of a prediction concerning themselves. It further finds that age is consistently measured as a covariate of technology adoption but has not been operationalized as a parameter within a predictive or adaptive model. We conclude by identifying a specific, current gap at the intersection of these threads: an explainable, age-aware system in which the teacher is both the predicted subject and the recipient of the explanation.

Keywords: explainable AI, learning analytics, teacher technology adoption, age-aware personalization, trait-aware systems

1 Introduction

The application of artificial intelligence to education has expanded substantially since 2021, driven in large part by advances in explainable machine learning and by the accessibility of large-scale behavioral and performance data through learning management systems. Two research threads relevant to teacher-facing systems have developed largely in parallel: explainable AI (XAI) methods applied to educational prediction, and research on teachers' adoption of AI and digital tools, including the role of age in that adoption. A third, smaller thread, concerning individual-difference-aware personalization in learning analytics, has emerged more recently still. This paper synthesizes peer-reviewed and archival literature from these threads, published between 2021 and 2026, to establish what is known, what methodological precedents exist, and where a specific gap remains open for further research.

The paper is organized thematically rather than chronologically. Section 2 describes the paper's approach and scope. Section 3 summarizes the established literature on teacher digital competence and AI adoption. Section 4 reviews explainable AI in educational analytics broadly. Section 5 examines the smaller body of work in which teachers interact directly with XAI systems. Section 6 reviews the literature on age as a variable in technology adoption. Section 7 reviews trait- and state-aware personalization as a structural precedent for age-aware system design. Section 8 synthesizes these threads into a specific, current research gap.

2 Approach and Scope

This is an informal, ad hoc synthesis rather than a formal systematic review: it does not follow a pre-registered protocol, dual-reviewer screening process, or formal quality-scoring instrument such as PRISMA. Literature was identified through iterative keyword searches conducted via Google Scholar and general web search, using combinations of "explainable AI," "teacher-facing," "learning analytics," "age," "technology adoption," "trait-aware," and "personalization," supplemented by forward and backward citation chaining from the most central papers in each thread (e.g., Khosravi et al. [4]; Borchers et al. [13]). Across the three threads, this process surfaced approximately ten directly relevant sources on teacher-facing XAI, eight on teacher age and technology adoption, and seven on trait- or state-aware personalization. This is not an exhaustive count: for comparison, Türkmen's formal systematic review of explainable AI in educational research, following the PICOS framework and database-driven search protocols, arrived at a corpus of 35 included studies after screening [6] — and even that formal review covers only one of the three threads synthesized here. The present synthesis was accordingly scoped for depth on the specific intersection under discussion rather than for exhaustive coverage of any one thread. This approach is appropriate for motivating a specific, bounded research question, but it does not support the exhaustiveness or risk-of-bias assessment of a full systematic review.

3 Teacher Digital Competence and AI Adoption: An Established Field

Research on teacher AI adoption has grown substantially since 2022, coinciding with the wider availability of generative AI tools. Ayanwale et al. examined teachers' readiness and intention to teach artificial intelligence in schools, establishing an early will-and-intention model of adoption prior to the generative AI wave [1]. Zhang et al. conducted a multigroup analysis of AI acceptance among pre-service teachers, finding acceptance patterns that varied by subgroup rather than uniformly across the sample [2]. More recently, Yi et al. modeled the link between AI usage intention and digital competence among college physical-education teachers using a moderated mediation design grounded in social cognitive theory and UTAUT [3]. Across this literature, adoption is modeled through attitudinal and intention-based constructs, self-efficacy, and perceived usefulness, using established technology-acceptance frameworks. The volume of recent synthesis work underscores how saturated this space now is: a systematic review of educational technology adoption in higher education screened 1,891 records down to 39 included studies spanning 2015–2024 [14], and a separate systematic review and network analysis specifically of the factors influencing teachers' technology acceptance drew on Scopus and Web of Science to identify 32 studies for in-depth analysis [15]. This body of work is mature and continues to expand rapidly; a proposal or study framed only as investigating teacher AI adoption or readiness would substantially overlap with existing research.

4 Explainable AI in Educational Analytics: Foundations and Growth

Khosravi et al. provided one of the field-defining position papers arguing that XAI in education has needs distinct from XAI in other domains, given the stakeholder diversity of students, teachers, and institutions, and the pedagogical stakes of algorithmic decisions [4]. Since that paper, SHAP, introduced by Lundberg and Lee [5], has become the dominant explanation method used in educational data mining.

A 2025 systematic review by Türkmen, following the PICOS framework across 35 studies published between 2019 and 2024, confirmed this predominance and found that XAI applications concentrate heavily on predicting and explaining student-level outcomes, including performance, engagement, and dropout risk [6].

Dropout and at-risk prediction is the most mature application area within this literature. Nagy and Molontay demonstrated interpretable dropout prediction using CatBoost together with SHAP and related tools, explicitly aiming to support personalized intervention design for university students, and reported that end users found some explanation formats, notably SHAP summary plots, difficult to interpret without training [7]. This finding is notable for the present synthesis: even in a mature, well-resourced application area, the usability of explanations for non-technical stakeholders remains an open, acknowledged problem, rather than a solved one.

5 Teacher-Facing XAI: Emerging but Still Validator-Centered

A smaller body of work brings teachers into direct contact with XAI systems, rather than treating them only as adopters of AI tools in general. Nazaretsky et al. developed and validated an instrument for measuring teachers' trust in AI-based educational technology, providing a reusable measurement tool for this specific population [8]. In a related study, Nazaretsky et al. co-designed a learning analytics tool for personalized instruction in the science classroom directly with teachers, following a participatory design process intended to align the system with teachers' own pedagogical reasoning [9]. Building on this line of work, Feldman-Maggor et al. proposed an XAI scheme for unsupervised machine learning, applied to a case study in which science teachers used an interactive dashboard, GrouPer, to examine clusters of students grouped by knowledge profile and to design differentiated follow-up activities [10]. More recently, Feldman-Maggor, Cukurova, Kent, and Alexandron examined how explainable AI shapes teachers' trust in and acceptance of AI EdTech recommendations, finding that domain-specific explanations were particularly effective in building teacher trust [16], reinforcing that this thread remains active into 2025 rather than settled by the earlier co-design studies.

Separately, Swamy et al. interviewed university educators directly about their trust in LIME and SHAP explanations of student-success predictions used for course design, finding that experts frequently disagreed with one another about which explanation method to trust, and that the same expert did not always favor the same method across different contexts [11].

This validator-centered pattern is echoed elsewhere in the broader XAI-in-education literature: Szymanski et al. found that teachers preferred relying on their own domain expertise over system-provided explanations when a distractor-generating AI performed poorly, again positioning the teacher as an expert evaluator of the system rather than its subject [17]. Across this thread, a consistent pattern holds: teachers are positioned as domain experts who interpret, validate, or act on explanations concerning someone else, typically students, rather than as the subject of the prediction themselves. This is a meaningful step beyond treating teachers only as passive recipients of AI tools, but it stops short of a configuration in which a system predicts and explains an outcome about the teacher's own practice or development, directed back to that teacher.

6 Age as a Variable in Technology Adoption: Measured, Not Modeled

Age is a frequently included covariate in teacher technology-adoption research, but findings across studies are inconsistent in direction and generally modest in magnitude. Baier-Mosch et al., studying 177 German teachers, found that age explained only 4 to 9 percent of variance across models of AI-related attitudes, knowledge, and perceived difficulty, with effects that diverged by outcome: positively associated with believing AI should be taught, negatively associated with positive affect toward learning AI content, and positively associated with perceived difficulty [12]. This pattern of small, multi-directional effects recurs across the broader adoption literature summarized in Section 3, where age is one covariate among several rather than a central object of study.

In every case reviewed, age is treated as an outcome-adjacent variable: something measured and reported alongside a model's findings, rather than a parameter embedded within the model's own predictive or adaptive logic. No study identified in this synthesis incorporates age as an interaction term, a stratification variable, or a moderator within a predictive system, despite the consistent finding that its effects are real, if modest, and vary by outcome rather than acting uniformly.

7 Trait- and State-Aware Personalization: A Structural Precedent

A structural precedent for closing the gap identified in Section 6 exists in recent work on individual-difference-aware learning analytics, developed for a different construct. Borchers et al. proposed a Trait-Aware Learning Analytics framework, presented at the LAK26 Learning Analytics and Knowledge Conference, arguing that learning analytics personalization has historically operated at the level of content or performance while overlooking relatively stable individual differences, such as personality traits [13]. The proposed framework embeds traits across the learning-analytics cycle as moderators that pre-specify for whom a given behavior or intervention is likely to have a larger or smaller effect, and as design primitives that inform the pacing, framing, and timing of support delivered to a given user.

This framework has been developed and illustrated specifically for personality constructs. It has not, to date, been extended to age, nor applied to a teacher-training or teacher professional-development context. Two further examples corroborate that this modeling move is an established, if still narrow, practice for personality rather than an isolated case. Takami et al. developed a personality-based tailored explainable recommendation approach for smart learning environments, using personality to shape both the recommendation and its accompanying explanation rather than reporting personality only as a correlate of outcomes [18]. By contrast, Matcha et al., in work Borchers et al. themselves cite as illustrative of the field's more common pattern, linked Big Five personality traits to MOOC learning-strategy profiles in a way the authors describe as descriptively valuable but not used to route supports or change system behavior [13]. Read together, these examples show that even within the personality-as-moderator literature, embedding the trait in the model's adaptive logic (as in Takami et al.) remains the exception rather than the rule (as in Matcha et al.) — underscoring that the equivalent step for age has essentially no precedent at all. Its existence nonetheless demonstrates that the methodological step required, treating an individual-difference variable as a moderator embedded in model logic rather than as a reported covariate, is established practice for at least one construct, and is a natural, if unexplored, extension for age.

8 Synthesis: The Open Gap

Read together, this literature supports four conclusions. First, teacher AI adoption and digital competence are extensively and continually studied, using mature technology-acceptance frameworks; this is not, on its own, an open area in need of further conceptual mapping. Second, explainable AI methods, particularly SHAP, are well established for predicting and explaining student-level outcomes, but usability challenges persist even in mature applications, and application to teacher-level outcomes remains limited. Third, where teachers do interact directly with XAI systems, the established pattern positions them as expert validators of predictions about students, a configuration explored across several recent studies [9, 10, 11], rather than as the subject of a prediction about their own practice or development. Fourth, age is a consistently measured but never modeled variable in the adoption literature, while a structural precedent for embedding an individual-difference construct as a moderator within a predictive system exists for personality traits but has not been extended to age or applied in a teacher-facing context [13].

The specific, current gap this synthesis identifies sits at the intersection of the third and fourth conclusions: no system identified in this literature explains a prediction about a teacher's own outcome, such as digital-skill development, back to that teacher, while treating age as a parameter shaping the prediction itself, rather than as a demographic label reported alongside it. Addressing this gap does not require inventing new explanation methods or new adoption theory; it requires adapting established teacher-facing XAI evaluation practice [8, 9, 10, 11] and established trait-aware modeling practice [13] to a configuration, teacher-as-subject, age-as-parameter, that current literature has not yet combined.

9 Conclusion

This paper has synthesized recent literature across three threads relevant to explainable, age-aware, teacher-facing learning analytics: teacher AI adoption, explainable AI in educational analytics, and trait-aware personalization. Each thread is individually active and, in places, mature. Their intersection, an explainable system that predicts and personalizes around a teacher's own developmental trajectory using age as a modeled parameter, remains open. Future work, including the author's own doctoral research, is positioned to address this intersection directly, building on the evaluation instruments, co-design practices, and modeling frameworks already established across the threads reviewed here, rather than treating any of them as a solved problem to be reinvented.

References

[1] M. A. Ayanwale, I. T. Sanusi, O. P. Adelana, K. D. Aruleba, and S. S. Oyelere, "Teachers' readiness and intention to teach artificial intelligence in schools," *Computers and Education: Artificial Intelligence*, vol. 3, Art. no. 100099, 2022.

[2] C. Zhang, J. Schießl, L. Plößl, F. Hofmann, and M. Gläser-Zikuda, "Acceptance of artificial intelligence among pre-service teachers: A multigroup analysis," *International Journal of Educational Technology in Higher Education*, vol. 20, no. 1, Art. no. 49, 2023.

[3] R. Yi, D. Liu, X. Sun, and B. Zhou, "Exploring the link between AI usage intention and digital competence among college PE teachers: A moderated mediation model based on SCT and UTAUT,"

PLOS ONE, 2025.

[4] H. Khosravi, S. Buckingham Shum, G. Chen, C. Conati, Y.-S. Tsai, J. Kay, S. Knight, R. Martinez-Maldonado, S. Sadiq, and D. Gašević, "Explainable artificial intelligence in education," *Computers and Education: Artificial Intelligence*, vol. 3, Art. no. 100074, 2022.

[5] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[6] G. Türkmen, "The review of studies on explainable artificial intelligence in educational research," *Journal of Educational Computing Research*, 2025.

[7] M. Nagy and R. Molontay, "Interpretable dropout prediction: Towards XAI-based personalized intervention," *International Journal of Artificial Intelligence in Education*, vol. 34, no. 2, pp. 274–300, 2024.

[8] T. Nazaretsky, M. Cukurova, and G. Alexandron, "An instrument for measuring teachers' trust in AI-based educational technology," in *Proc. LAK22: 12th Int. Learning Analytics and Knowledge Conf.*, 2022.

[9] T. Nazaretsky, C. Bar, M. Walter, and G. Alexandron, "Empowering teachers with AI: Co-designing a learning analytics tool for personalized instruction in the science classroom," in *Proc. LAK22: 12th Int. Learning Analytics and Knowledge Conf.*, 2022, doi: 10.1145/3506860.3506861.

[10] Y. Feldman-Maggor, T. Nazaretsky, and G. Alexandron, "Explainable AI for unsupervised machine learning: A proposed scheme applied to a case study with science teachers," in *Proc. 16th Int. Conf. Computer Supported Education (CSEDU)*, vol. 1, 2024, pp. 436–444, doi: 10.5220/0012687000003693.

[11] V. Swamy, S. Du, M. Marras, and T. Käser, "Trusting the explainers: Teacher validation of explainable artificial intelligence for course design," in *Proc. LAK23: 13th Int. Learning Analytics and Knowledge Conf.*, Arlington, TX, USA, 2023, doi: 10.1145/3576050.3576147.

[12] F. Baier-Mosch, L. Schulze-Vorberg, and J. Schneider, "The relationship between teachers' age, gender and their will and skill to teach artificial intelligence," *Technology, Knowledge and Learning*, 2026.

[13] C. Borchers, H. Deininger, and Z. A. Pardos, "Toward trait-aware learning analytics," in *Proc. LAK26: 16th Int. Learning Analytics and Knowledge Conf.*, Bergen, Norway, 2026.

[14] J. Feng, B. Yu, W. H. Tan, Z. Dai, and Z. Li, "Key factors influencing educational technology adoption in higher education: A systematic review," *PLOS Digital Health*, 2025, doi: 10.1371/journal.pdig.0000764.

[15] E. G. Galimova, A. Yu. Oborsky, M. A. Khvatova, D. V. Astakhov, E. V. Orlova, and I. S. Andryushchenko, "Mapping the interconnections: A systematic review and network analysis of factors influencing teachers' technology acceptance," *Frontiers in Education*, vol. 9, Art. no. 1436724, 2024, doi: 10.3389/feduc.2024.1436724.

[16] Y. Feldman-Maggor, M. Cukurova, C. Kent, and G. Alexandron, "The impact of explainable AI on teachers' trust and acceptance of AI EdTech recommendations: The power of domain-specific explanations," *International Journal of Artificial Intelligence in Education*, vol. 35, no. 5, pp. 2889–2922, 2025, doi: 10.1007/s40593-025-00486-6.

[17] M. Szymanski, J. Ooge, R. De Croon, V. Vanden Abeele, and K. Verbert, "Feedback, control, or explanations? Supporting teachers with steerable distractor-generating AI," in *Proc. LAK24: 14th Int. Learning Analytics and Knowledge Conf.*, Kyoto, Japan, 2024, pp. 690–700, doi: 10.1145/3636555.3636933.

[18] K. Takami, B. Flanagan, Y. Dai, and H. Ogata, "Personality-based tailored explainable recommendation for trustworthy smart learning system in the age of artificial intelligence," *Smart Learning Environments*, vol. 10, Art. no. 65, 2023, doi: 10.1186/s40561-023-00282-6.